

Planning with Uncertainty

Block 2 Day 4

Semester 1, 2026–2027 — Teaching Week 4

Session Overview

Session objective: Move from utility and decision networks to sequential planning with Markov decision processes and the limits of full observability.

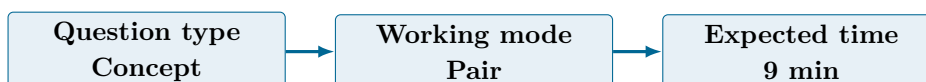
Timing target (min): 71

Topic anchors: utility; expected utility; decision network; markov decision process; value iteration; policy iteration

Padlet board: <https://padlet.com/qmul/b4-d4-4yoefwupfn46p8ph>

Exercises

Q1 Starter Classification



Prompt

In an emergency drone-routing problem, classify these items as feature, preference, or utility term: battery level, wind speed, avoid school zone, short travel time, crash penalty, and patient arrival bonus. Then explain why mixing these categories causes planning mistakes.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 9 min
- **Deliverable:** concept table
- **Assessment focus:** separate features, preferences, and numeric utility terms
- **Expected length:** 6 labels + 1 explanation

How to submit

Post one pair Padlet comment as a six-row table plus one short explanation.

Discussion in class

Which item is easiest to classify wrongly?

Why this helps

Useful for questions that ask students to separate the world description from the planner's objective.

References

- Russell & Norvig (2021) AIMA 4e, Ch.16-17
- Poole & Mackworth (2023) 3e, Ch.12, pp.517-582

Q2 Structured Problem

Question type
Problem

Working mode
Individual

Expected time
11 min

Prompt

A delivery agent has two route choices. Fast route: with probability 0.6 it arrives on time for utility +12, and with probability 0.4 it suffers severe delay for utility -10. Safe route: with probability 0.85 it arrives on time for utility +8, and with probability 0.15 it suffers mild delay for utility +1. Compute expected utility for both actions and choose the better one.

Formula reminder

$$EU(a) = \sum_s P(s \mid e) U(s, a)$$

Use this when the task asks you to compare actions under uncertainty.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 11 min
- **Deliverable:** expected utility computation
- **Assessment focus:** compute expected utility and choose the better action
- **Expected length:** 4 calculation lines

How to submit

Post your own Padlet comment showing both expected-utility calculations and the chosen action.

Discussion in class

Why can the route with the larger best-case reward still be the worse decision?

Why this helps

Direct practice for expected-utility calculations without duplicating the exam numbers.

References

- Russell & Norvig (2021) AIMA 4e, Ch.16-17
- Poole & Mackworth (2023) 3e, Ch.12, pp.517-582

Q3 Structured Problem

Question type
Problem

Working mode
Individual

Expected time
12 min

Prompt

For each scenario, choose the most suitable planning model and explain why: (1) a one-off hiring decision with uncertain outcomes, (2) a warehouse robot making repeated fully observed navigation choices, and (3) a rescue robot acting repeatedly when the true hazard level is only partly visible. Choose from decision network, Markov decision process (MDP), or partially observable Markov decision process (POMDP).

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** model-choice explanation
- **Assessment focus:** choose between a decision network, a Markov decision process, and a POMDP-style view
- **Expected length:** 3 short cases

How to submit

Post your own Padlet comment with three choices and one short reason each.

Discussion in class

Which feature forces you away from an ordinary MDP?

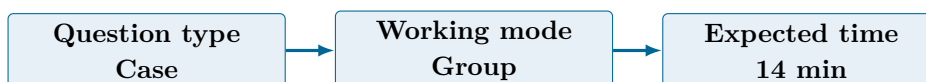
Why this helps

Useful for model-selection questions that test when partial observability changes the planning model.

References

- Russell & Norvig (2021) AIMA 4e, Ch.16-17
- Poole & Mackworth (2023) 3e, Ch.12, pp.517-582

Q4 Collaborative Mini-Case



Prompt

Design a hospital-delivery robot planning setup that shows the difference between a one-off choice and a sequential choice. Include state, action, uncertainty, utility, and one reason the system might later need a belief-based view rather than a plain MDP.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 14 min
- **Deliverable:** sequential planning sketch
- **Assessment focus:** design a small planning setup that clearly becomes sequential
- **Expected length:** 5 bullets + 1 note

How to submit

Post one group Padlet comment with the planning setup and one note on where uncertainty changes the choice.

Discussion in class

Which part of your design makes the problem sequential instead of one-shot?

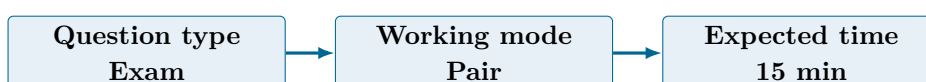
Why this helps

Supports case questions on moving from simple decision analysis to sequential planning.

References

- Russell & Norvig (2021) AIMA 4e, Ch.16-17
- Poole & Mackworth (2023) 3e, Ch.12, pp.517-582

Q5 Exam-Style Constructed Response



Prompt

Explain the difference between planning in a Markov decision process with a known model and learning by reinforcement in the same environment. Then add one short note on why partial observability makes planning harder.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 15 min
- **Deliverable:** written argument
- **Assessment focus:** compare planning in a Markov decision process with learning by reinforcement
- **Expected length:** 180-240 words

How to submit

Post one pair Padlet comment with a clear comparison, one reason the known model matters, and one note on partial observability.

Discussion in class

Why is planning in an MDP not just “reinforcement learning done more carefully”?

Why this helps

Strengthens comparison of planning and learning methods in a new setting.

References

- Russell & Norvig (2021) AIMA 4e, Ch.16-17
- Poole & Mackworth (2023) 3e, Ch.12, pp.517-582

Q6 Stretch Extension

Question type
Stretch



Working mode
Group



Expected time
10 min

Prompt

Build a short safety checklist for uncertain planning in a high-risk domain such as health-care delivery, emergency response, or cybersecurity response.

Task details

- **Type:** stretch
- **Mode:** group
- **Time:** 10 min
- **Deliverable:** safety checklist
- **Assessment focus:** turn uncertain planning into operational controls
- **Expected length:** 4-5 checklist items

How to submit

Post one group Padlet comment with four or five controls and say which one should be enforced first.

Discussion in class

Which safety check protects against the worst planning mistake?

Why this helps

Good extension for questions that ask students to move from planning theory to safe deployment.

References

- Russell & Norvig (2021) AIMA 4e, Ch.16-17
- Poole & Mackworth (2023) 3e, Ch.12, pp.517-582

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1 and Q5 are pair tasks. Q4 and Q6 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which utility calculation changed the decision most?
- What makes a planning problem sequential rather than one-off?
- When does partial observability force you to change the planning model completely?

Exam Transfer Note (Day-Level)

Maps to exam questions on utility, expected utility, decision networks, Markov decision processes, and planning under limited observability.